

1. Administrative & Study Identification

Full Study Title: Efficacy of FDC Regimen of Dapagliflozin/Metformin Compared to Co-administered Dual Therapy on Glycemic Control, Satisfaction and Adherence in Chinese Patients With T2DM
Short Title/Acronym: Dapagliflozin/Metformin FDC vs Dual Therapy in T2DM
Protocol Number: D1690L00149 **NCT Number:** NCT06327815
Trial Status: Completed **Sponsor Name:** AstraZeneca **Investigational Product:** Dapagliflozin 10 mg / Metformin hydrochloride extended-release (XR) 1000 mg Fixed-Dose Combination (FDC) tablet (Xigduo XR)
Phase of Development: Phase IV **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate whether the efficacy of a fixed-dose combination (FDC) regimen of dapagliflozin/metformin XR is non-inferior to co-administered dual therapy on glycemic control (change in HbA1c) after 24 weeks of treatment in newly diagnosed type 2 diabetes mellitus (T2DM) patients. **Secondary Objectives:** To evaluate the proportion of patients achieving HbA1c < 7.0%, absolute change in fasting plasma glucose (FPG) and postprandial glucose (PPG) from baseline to week 24, and differences in patient satisfaction, medication adherence, and quality of life between the two groups.

2.2 Background & Rationale To address medication adherence and optimize glycemic control in newly diagnosed T2DM patients. Sodium-glucose cotransporter 2 (SGLT2) inhibitors and metformin are widely recommended as foundational therapies. Although bioequivalence has been established for the dapagliflozin/metformin XR FDC compared to separate co-administration, it remains uncertain whether the FDC's efficacy and patient adherence are comparable in a real-world clinical setting for the Chinese population. Fixed-dose combination therapy may significantly improve patient adherence and satisfaction compared to separate multi-pill regimens.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Co-administered dual therapy) **Blinding:** Open-label **Randomization:** 24-Week, Multicentre, Randomized, Parallel, Interventional, Non-inferiority design

3.2 Planned Enrollment & Duration Trial Status: Completed **Target Enrollment:** 633 patients **Number of Sites:** 35 research centers in China **Estimated Study Start:** March 2024 **Estimated Primary Completion:** 24 weeks post-randomization

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 and ≤ 80 years at the time of signing informed consent. 2. Newly diagnosed T2DM (WHO diagnostic criteria 1999) ≤ 1 year and medicine treatment naïve. 3. HbA1c 7.5% – 10.0% at screening by local laboratory and pre-randomization visit (central laboratory). 4. BMI ≥ 19 and ≤ 40 kg/m² at screening. 5. Capable of giving signed informed consent and complying with study requirements.

4.2 Exclusion Criteria 1. Congestive heart failure NYHA classes III or IV or major cardiovascular events within 6 months before screening (e.g., myocardial infarction, coronary angioplasty, valvular disease, unstable angina, TIA, or stroke). 2. Estimated glomerular filtration rate (eGFR) < 45 mL/min/1.73 m². 3. Clinically apparent hepatobiliary disease, including chronic active hepatitis or severe hepatic insufficiency (ALT or AST > 3x ULN, or total bilirubin > 34.2 μ mol/L [> 2 mg/dL]). 4. Diagnosis or history of acute metabolic diabetic complications (e.g., ketoacidosis, hyperglycemic hyperosmolar state, or diabetes insipidus) within the past 6 months. 5. Diagnosis of T1DM, diabetes resulting from pancreatic injury or secondary forms of diabetes (e.g., acromegaly or Cushing's syndrome). 6. Current pregnancy (confirmed by positive test) or breastfeeding. 7. Participation in any other study that included drug treatment during the past 3 months. 8. Known hypersensitivity to Dapagliflozin/Metformin or any product excipients. 9. Chronic pancreatitis within past 6 months or idiopathic acute pancreatitis within past 4 weeks. 10. Gastrointestinal disease including gastroenterostomy, enterectomy, roemheld syndrome, severe hernia, intestinal obstruction, or intestinal ulcer within past 6 months. 11. Genetic galactose intolerance, lapp lactase deficiency and glucose-galactose malabsorption. 12. Medullary thyroid carcinoma within past 5 years. 13. Organ transplant or AIDS within the past 6 months. 14. Alcohol abuse or illegal drug abuse within the past 12 months. 15. Laser treatment for proliferative retinopathy within 6 months. 16. Stress condition (e.g., surgery, serious trauma) within past 6 months, or planned surgery during study period. 17. Chronic oxygen deficiency diseases (e.g., pulmonary emphysema, pulmonary heart disease) within past 6 months.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Investigational Arm:** Fixed-Dose Combination (FDC) tablet of 10 mg dapagliflozin

and 1000 mg metformin XR, taken once daily. - **Control Arm:** Co-administered 10 mg dapagliflozin tablet and 1000 mg metformin XR tablet, taken once daily. **Route of Administration:** Oral **Duration of Treatment:** 24 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant therapies for hypertension, dyslipidemia, and other co-morbidities as deemed appropriate by the investigator. **Prohibited:** Concurrent use of other anti-diabetic medications or strong CYP inducers/inhibitors that may significantly alter the pharmacokinetics of dapagliflozin or metformin.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, BMI, HbA1c, eGFR, ECG
Baseline	Day 0	Randomization, First Dose, Treatment Satisfaction Baseline Questionnaire
Interim	Week 12	Safety Labs, Efficacy Assessment, Quality of Life Questionnaire
End of Study	Week 24	Final HbA1c/FPG/PPG Labs, Adherence & Satisfaction Scales, AE/SAE Review, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in HbA1c level from baseline after 24 weeks of treatment. To demonstrate non-inferiority of Dapa/Met FDC regimen when compared with Dapa + Met co-administered. **Measurement Method:** Central Laboratory Analysis.

7.2 Secondary & Exploratory Endpoints - Proportion of patients achieving HbA1c target of < 7.0% from baseline to week 24. - Absolute change in fasting plasma glucose (FPG) from baseline to week 24. - Absolute change in postprandial glucose (PPG) from baseline to week 24. - Difference in patient satisfaction measured using the diabetes treatment satisfaction questionnaire. - Patient medication adherence measured using the adherence to refills and medications scale. - Change from baseline in diabetes quality of life questionnaire score.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical evaluations, laboratory assessments (routine blood and urine tests), and patient reporting. **Serious Adverse Events (SAEs):** Any untoward medical consequence resulting in death, life-threatening scenarios, or requiring inpatient hospitalization will be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal and external safety monitoring conforming to GCP standards.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) set for primary effectiveness outcomes; Per-Protocol (PP) set for non-inferiority validation; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Target sample of \$N=633\$ patients powered to test the non-inferiority margin for the primary endpoint (change in HbA1c). **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) or Multiple Imputation approaches will be utilized for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote clinical monitoring visits to ensure protocol compliance and data integrity. **Audit Trail:** eCRF entry tracking, laboratory data clarification forms, and standard quality audit mechanisms for protocol deviations.

11. Study Identifiers & Governance

Primary ID: D1690L00149 **Registry Number:** NCT06327815 **Posting ID:** 744230944806956143 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Efficacy of FDC Regimen of Dapagliflozin/Metformin Compared to Co-administered Dual Therapy on Glycemic Control, Satisfaction and Adherence in Chinese Patients With T2DM **Official Regulatory Title:** Efficacy of FDC Regimen of Dapagliflozin/Metformin Compared to Co-administered Dual Therapy on Glycemic Control, Satisfaction and Adherence in Chinese Patients With T2DM: A 24-Week, Multicentre, Randomized, Parallel, Interventional, Non-inferiority, Open-label Study

12. Clinical Framework (T2DM Specific)

Primary Condition: Type 2 Diabetes Mellitus (T2DM) **Diagnosis Threshold:** HbA1c 7.5% - 10.0% **Medical Methodology:** Fixed-Dose Combination Therapy vs. Separate Co-administration **Optimization Target:** Target glycemic control (HbA1c < 7.0%) and improved medication adherence

13. Study Procedures & Methodology

Management Pathway: SGLT2 inhibitor + Metformin foundational antidiabetic therapy **Education Protocol (HCP):** Protocol specific dosing and adherence-tracking training **Education Protocol (Patient):** Guidance on glucose monitoring, FDC adherence, and

metabolic disease management **Quality Audit Mechanism:** Application of the Adherence to Refills and Medications Scale and the Diabetes Treatment Satisfaction Questionnaire

14. Observation & Follow-Up Schedule

Total Duration: 24 weeks of active treatment **Onsite Visit Cadence:** Baseline, Week 12, Week 24 **Remote Monitoring Frequency:** Routine site-level clinical monitoring as per GCP **Checklist Review Interval:** End of study adherence and quality-of-life assessments at Week 12 and Week 24

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				